

VALLEY HEAD 1 SSW, AL, USA

WMO: 720011

Lat: 34.5653N

Lon: 85.6171W

Elev: 1020

StdP: 14.16

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63862

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 15.2	(c) 19.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 19.8	(c) 91.2	(d) N/A	(e) 89.2	(f) N/A	(g) 87.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Externe Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		Extreme Max WB	
(a) N/A	(b) N/A	(c) N/A	(e) 9.4	(f) 93.8	(g) 4.7	(h) 3.3	(i) 6.0	(j) 96.1	(k) 3.3	(l) 98.0	(m) 0.7	(n) 99.9	(o) -2.7	(p) 102.3		
			(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.0	38.9	41.9	50.8	58.8	66.9	74.1	76.2	76.4	71.1	59.6	48.4	44.0
	DBStd	15.37	10.81	9.65	9.19	7.60	6.43	3.68	3.41	3.88	5.60	8.44	8.44	9.71
	HDD50	1139	374	259	105	17	0	0	0	0	16	127	241	
	HDD65	3543	810	647	443	209	60	1	0	16	208	499	650	
	CDD50	4438	29	33	130	281	524	724	813	819	634	314	81	56
	CDD65	1366	0	0	3	23	119	275	348	354	200	41	2	1
	CDH74	11066	0	1	57	260	999	2334	2700	2895	1510	301	9	0
	CDH80	3791	0	0	3	18	222	857	984	1152	498	56	1	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	56.7	4.9	4.5	6.5	5.0	4.8	4.6	5.6	3.6	4.6	3.2	5.0	4.4
	PrecMax	69.8	9.6	13.6	15.9	10.9	11.3	10.0	12.8	6.9	11.0	9.9	10.3	10.5
	PrecMin	41.7	1.3	0.6	1.9	1.2	1.1	0.6	0.9	0.6	0.8	0.1	1.3	0.8
	PrecStd	8.3	2.0	2.3	3.3	2.3	2.4	2.4	2.8	1.7	3.0	2.2	2.1	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.7	71.0	79.7	82.3	87.2	91.8	92.4	95.3	91.4	85.0	74.5	69.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	63.7	66.4	75.1	79.7	84.9	89.7	89.9	91.9	88.6	80.6	69.7	67.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	60.2	63.1	71.0	77.3	82.7	87.6	87.9	89.1	86.0	77.3	66.8	63.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	56.5	59.4	67.0	74.2	80.2	85.5	85.8	86.6	82.6	73.9	63.8	59.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	19.3	21.5	22.6	24.2	21.8	21.3	19.0	19.8	20.5	22.7	23.5	18.5
		MCDBR	20.7	25.8	26.8	27.1	25.2	24.7	21.6	23.2	24.7	24.5	25.5	19.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.314	0.319	0.356	0.393	0.445	0.481	0.533	0.530	0.424	0.372	0.337	0.314	
	taud	2.508	2.490	2.412	2.295	2.191	2.124	1.995	1.991	2.280	2.413	2.458	2.528	
	Ebn at Noon	284	295	291	283	268	258	243	242	265	272	272	278	
	Edh at Noon	26	30	35	42	47	50	56	56	40	32	28	24	
All-Sky Solar Radiation	RadAvg	788	954	1311	1702	1871	1967	1865	1739	1496	1230	893	669	
	RadStd	57	124	121	111	145	149	121	93	139	157	102	67	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+261	+189	

Nomenclature: See separate page